

Augmenting Murals: Creating Playful AR Experiences for Wall Art Exhibitions

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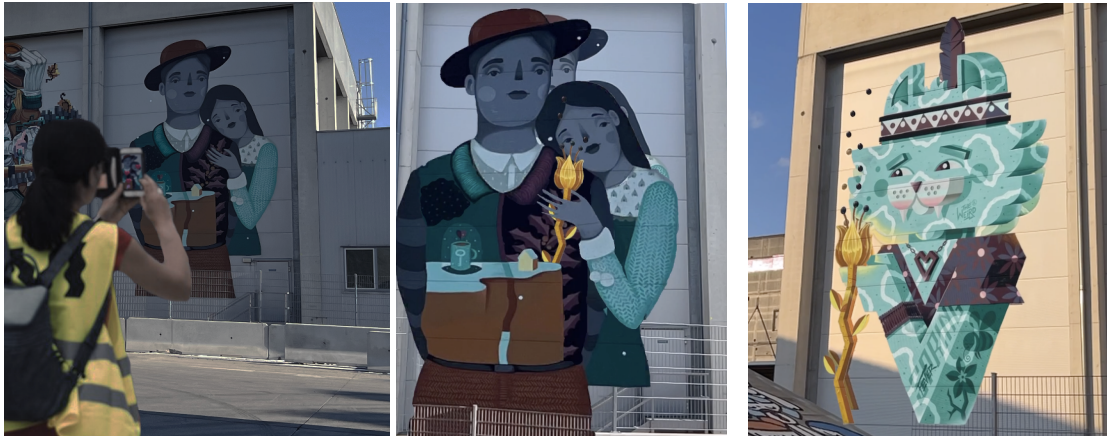


Figure 1: A visitor using the mobile AR application while pointing the smartphone onto a mural (left) and augmented murals (middle and right) that can interact with each other. The mural on the right brings the golden flower to the mural displayed in the middle. Additionally, the murals are animated, enabling them to blink with their eyes, move their arms and change their position.

ABSTRACT

Outdoor art exhibitions in natural settings offer a popular environment for people to engage with art. With the rapid advancements in augmented reality and mixed reality, immersive technologies and digital augmentations are transforming how people experience art by incorporating animations and interactive elements. However, marker-based augmented reality (AR) approaches, though commonly used, present limitations in large outdoor exhibitions, including labour-intensive setups and restricted exploration freedom for users. To address these challenges, we propose a markerless AR approach utilizing *Geospatial* tracking, which appears promising for enhancing large-scale outdoor art exhibitions. In this paper, we present the development of a mobile AR application designed to augment murals with dynamic animations and interactive features. We explore various interaction concepts, demonstrating how this technology can be effectively utilized to create innovative, user-centered experiences in outdoor art settings.

Index Terms: outdoor art exhibitions, visualization, augmented reality, murals

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Applied computing—Arts and humanities—Media arts

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1 INTRODUCTION AND BACKGROUND

Outdoor art exhibitions become increasingly popular as the natural setting enhances the experience of viewing art [13]. *Mural Harbor*¹, for example, is an internationally renowned outdoor art exhibition in the Linz harbour area in Austria, in which graffiti/murals have been painted on entire facades of warehouses since 2014. These are spread across the entire site, painted in all possible sizes by different artists from around the globe and can be viewed as part of a paid guided tour [12]. Due to the progress of immersive technologies such as AR and mixed reality (MR), their popularity and social acceptance have increased [27, 16]. As research has shown, adding a virtual dimension to exhibition spaces leads to an enhanced visitor experience [29, 19]. Offering an AR mobile app that allows visitors to experience the murals in more dynamic or interactive ways [28] could result in a more innovative and engaging exhibition. For example, using *Artivive*², several mural paintings on abandoned buildings were brought to life by placing virtual objects. These would tell the story of the buildings and thus re-cultivate them [29, 1]. Another example shows how the famous painting *Along the river during the Qingming Festival*³ was recreated and animated using virtual objects and sound effects which highlighted or supplemented individual elements and information [6].

Overlaying interactive digital elements onto the physical artworks ties layers of animation, storytelling, or contextual information to the art [25]. Projection mapping, for example, transforms static murals into dynamic displays by projecting animations and interactive visuals onto surfaces [20, 11]. Additionally, beacons can enhance the visitor experience by triggering localized content,

¹<https://muralharbor.at/>

²A smartphone app using marker-based AR in which artists can upload their AR art and set their (static) artwork as a marker [1].

³<https://theme.npm.edu.tw/exh105/RiverQingming/en/index.html>



Figure 2: Comparison of different variations of an abstract wall mural: 3D model (left), 2D animation (center), and 2D parallax effect (right).

such as audio guides or artist commentaries, tailored to specific parts of an exhibition [24]. Together, these tools can empower artists and curators to push the boundaries of traditional art presentations, creating engaging, multi-sensory experiences for diverse audiences [26].

Current research shows that developing marker-based tracking gives the highest tracking accuracy. However, this requires the marker to be sufficiently complex, clearly visible, and not viewed from too close, too far away or too steep in angle [4]. Another prerequisite – particularly in outdoor areas – is the marker being placed prominently enough in the environment and is recognized as such by users without requiring explicit instructions. By doing so, murals can tell stories both with their design and their position in the environment [29]. Markerless tracking technologies are therefore a promising technology to create AR experiences in public spaces. While they often require sensors that are not installed in all devices or might be based on technology that is too inaccurate (such as traditional location-based systems) [4], a recently developed system called *Geospatial* tracking can overcome the challenges of marker-based systems, as it does not need expensive equipment [21, 8].

Hence, we propose an AR system leveraging *Geospatial* tracking [8]. In this system, the visitor's location is first determined using GPS. Then, the accuracy of location-tracking is further refined using the camera data of the visitor's smartphone. This is done by sending the data to a Google server. Next, the data is compared with Google's *Street View* data using a machine learning model. Therefore, this system offers a markerless solution, allowing free placement of the virtual objects, while still maintaining a fairly high level of accuracy if *Street View* coverage is sufficient [8].

Using this approach, AR experiences can be designed to persist at a specific location regardless of minor environmental changes, as they remain functional even if markers are overgrown, painted over, or entirely absent [29]. This implies that adopting a markerless approach also simplifies the process by removing the need for designing and placing physical markers [29, 4]. Instead, virtual objects are given a specific location using WGS84⁴ values which store longitude, latitude and altitude [10, 14]. The development of such AR experiences is also expected to be more accessible, as popular game engines such as *Unity* combined with frameworks (for example *AR-Foundation* [7]) enable deployment across a wide range of devices. Using *Unity* together with the *Geospatial Editor* and *Cesium* plugin also enables the automatic generation of position values by placing

virtual objects on a 3D map inside *Unity* [3, 9]. This WYSIWYG⁵ approach could greatly benefit less experienced developers or artists to have easier access to AR development. As a result, this opens up new possibilities in placement and the narrative, since virtual objects can now be placed in locations that would not originally have been suitable markers. Therefore, we describe the implementation of a markerless mobile AR system using the *Geospatial Creator* for exhibitions in large spaces and propose a study design to evaluate it using mixed methods.

2 AUGMENTING MURALS USING MARKERLESS AR

In the following, we briefly describe the implementation of a markerless approach for digitally augmenting murals based on *Mural Harbor* in Linz, Austria. *Mural Harbor* is an outdoor art exhibition located in the harbour area, making *Geospatial* tracking a great fit since it is restricted to outdoor use only. As mentioned, the implementation using *Geospatial* tracking offers new features and experiences [8]. For example, the starting area of the tour can be virtually supplemented with additional elements such as floating info-graphics and murals that can be interacted with – turning the starting area into a tutorial playground for the upcoming tour. This approach has the potential to suggest visitors to look around the area in a more detailed fashion, rewarding them with an interaction/reaction if they find virtual counterparts of the murals or standalone virtual objects that enhance the real environment. Consequently, this creates novel and playful ways of experiencing outdoor art exhibitions similar to a sandbox game, providing the visitors with a high degree of freedom usually without any predetermined goal [23].

2.1 Prototype Development

Because *Geospatial* tracking relies on comparing the camera data with the *Street View* images from Google, a connection between *Unity* and a Google server is necessary. Specifically, the *Google Map Tiles API*, the *ARCore Geospatial API* and the *Places API* are required. Once these are activated, the API key from the *Google Cloud Server* can be entered into the *Geospatial Editor* plugin inside *Unity*. After completing the setup of the *Cesium* plugin and defining the preferred location, virtual objects can be placed inside the *Unity* scene.

⁴World Geodetic System 1984, which is a uniform reference system as the basis for position data on Earth.

⁵What You See Is What You Get

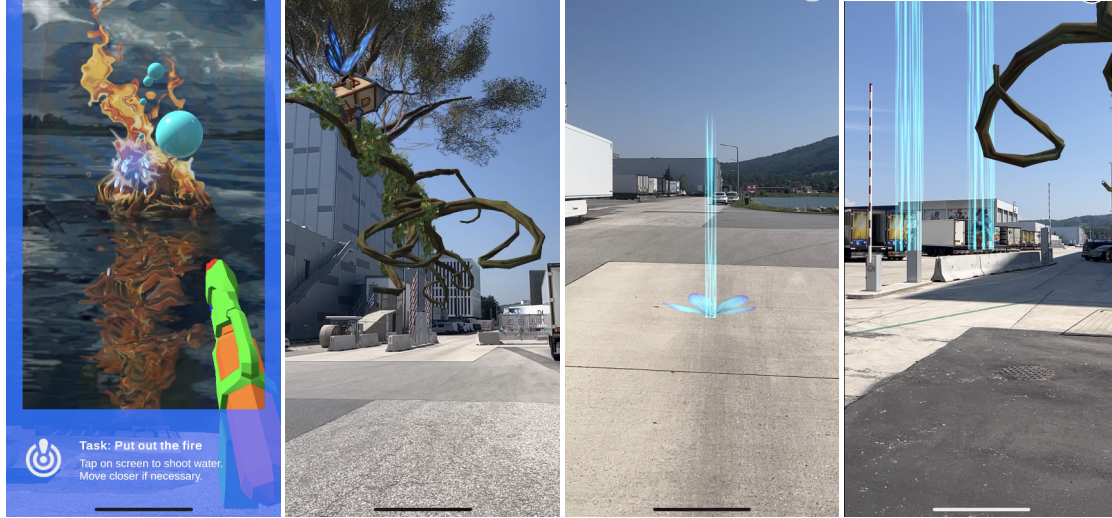


Figure 3: View of several AR experiences of the prototype such as the mini-game where the user extinguishes a fire mural by aiming and shooting a virtual water gun at the artwork (left), the decorative virtual tree without a mural (center left), a waypoint projected onto the ground serving as navigational aid (center right) and the landmarks indicating a virtual mural (right).

2.1.1 Preparations

We brainstormed ideas to digitally augment murals and design engaging AR experiences. Simultaneously, we captured the murals using photographs which could be further processed. Both editing as a raster graphic and converting to a vector graphic were evaluated. However, we chose the first approach as it promised to be more simple and result in more detailed, virtual murals. In doing so, we used image editing software, such as *Adobe Photoshop*, to mask the desired areas and split up the mural into various layers. Those could then be used to animate specific parts (e.g., eyes, arms) of the mural. Inspired by research from avatar design and virtual agents [15, 17], we aimed to animate the murals. As the animation would also reveal parts which were not initially visible, it was sometimes necessary to reconstruct those areas. For example, Fig. 1 illustrates a mural of a couple with animated facial expressions, where the woman's arm moves to stroke the man's chest.

Two approaches were taken to animate parts of the murals. The first approach targeted frame-by-frame animation, in which we used video manipulation software, such as *Adobe After Effects*, to achieve transformations. Fig. 3 shows a mural of a campfire site which was animated/morphed using a displacement effect. After completion, the animation was imported in *Unity* and replaced the corresponding static layer.

For the second approach, the animation was created using *Unity*'s built-in functionality. This has the advantage of higher framerate motion and allows more flexibility in the development process. An example of this approach is the woman's arm movement seen in Fig. 1.

2.1.2 Virtual Object Types

In general, we chose three different types of virtual murals for this prototype: 3D, 2D parallax and 2D animated. For example, an abstract mural seen in Fig. 2 was implemented in all three types to prove their different advantages or disadvantages.

As already mentioned, we not only used virtual murals but also placed virtual objects which were not originally found in the environment. By adding these, they have the potential to influence/reinforce the narrative of the environment [29]. For example, to change the narrative of the harbour environment, being relatively

plantless in reality, a giant 3D tree was implemented in the prototype, as seen in Fig. 3. To achieve the desired effect, the *Tree Animate*⁶ model, the *Aaron project – Roots*⁷ model, and the *Fantasy House*⁸ model, were used from *Sketchfab*.

2.1.3 Interactions

When designing mobile AR experiences, tap interactions were selected as the primary input method as they are easy to use and common for smartphone usage [18]. Typically, if an interaction with a mural is detected using *Unity Raycasts*, a specific mural action and its interaction animation will be activated. The location where the mural is touched and the direction to the player are needed to ensure that the spawned objects are rotated towards the player. Since the virtual objects inherit a specific *WGS84* location in 3D space, we further explored interactions between murals, such as a lion mural dynamically delivering a golden flower to the couple mural when the user initiates interactions (see Fig. 1). This also serves as an example of micro-storytelling that could span the exhibition area: Murals could not only tell their story based on their design and position but also based on the interaction with surrounding murals and their specific location. This approach implies that murals have spatial awareness to enhance their (environmental) storytelling.

Fig. 3 illustrates another integrated feature: a fire-themed mural that users must extinguish by aiming and shooting a virtual water gun as part of a mini-game. This is achieved through repetitive tapping on the phone screen, however, the mini-game can only be completed when the user gets close enough to the mural as the virtual water droplets would not hit the mural from a greater distance. Therefore, the distance to both real and virtual objects is important when the visitor has to walk closer to the mural while looking out for objects that could occlude the mural. This adds an interactive and engaging element to the experience while showcasing a possible playful approach to AR exhibitions that incorporates the

⁶<https://sketchfab.com/3d-models/tree-animate-f0f9eb5e6c104bbb8e1f41c97019e6f2>

⁷<https://sketchfab.com/3d-models/aaron-project-roots-3ca6aba2e3d9453cb0e7bb651331bdf3>

⁸<https://sketchfab.com/3d-models/fantasy-house-ed9c83a3f88a4b5682a40e1180ab91e0>

surroundings. We used the *Water Gun*⁹ model from *Sketchfab*.

We also plan to find out if there is an impact on the visitor's engagement by experimenting with the narrative effect of user interactions on environmental storytelling. Specifically, by changing the environment to a colder one by adding virtual, falling snow when the visitor extinguishes the fire and thus completes the mini-game.

2.1.4 Navigation

As the visitors should not only be able to explore the site freely, we also considered elements which would guide the user. After some initial brainstorming, two elements were selected: Landmarks and waypoints. Landmarks should be visible from a distance and thus be able to draw the visitor's attention to a virtual mural from a greater distance (as seen in Fig. 3). On the other hand, waypoints should be a playful element, leading step by step to the next location, with a new waypoint appearing whenever the previous one is reached. To create the waypoint, the *Flower*¹⁰ model was used from *Sketchfab*.

2.2 Planned Study Design

Our prototype is designed for visiting outdoor art exhibitions using a smartphone. To evaluate the usability, functionality, and user experience, we plan to conduct an in-situ study using mixed methods. For this, we aim to simulate an unguided tour at *Mural Harbor* with a predefined starting point and route allowing participants to explore the murals freely. We will choose the route based on the murals that are digitally augmented and animated. During the run, the participants will be allowed to explore the area freely in a sense that they can return to already visited murals and locations. The experimenter is planned to follow the participants to support them in case of questions or problems. The participants will be asked to express their thoughts during a study by thinking aloud, which will be recorded using a smartphone. After finishing the tour, which is planned to take about 45 minutes, a semi-structured interview will be conducted. Quantitative data will also be collected at the end of the study using questionnaires such as the System Usability Scale [2] or User Experience Questionnaire [22]. The text responses to the open-ended questions will be analyzed following a thematic content analysis [5]. After the study, the audio recordings will be transcribed and manually supplemented with statements, context and, where necessary, the reason for pauses.

3 SUMMARY

In this proposal, we describe an AR application using *Geospatial* tracking in outdoor art exhibitions, focusing on its potential for enhancing user interaction and engagement. In contrast to marker-based tracking, our approach is a markerless technology that allows visitors to interact with virtual elements naturally and intuitively, without requiring them to locate or scan specific markers. Consequently, we document the development of the mobile AR prototype using *Geospatial Creator* in the *Unity* game engine. *Geospatial* tracking enabled innovative features such as navigation aids (e.g., landmarks or waypoints), animated interactions between murals (see Fig. 1), mini-games affecting the virtual environment, and decorative virtual elements like trees and houses (see Fig. 3), which could support the exhibition narrative. We hypothesize that our AR application is promising to increase user experiences in art exhibitions and create engaging and playful AR experiences. Based on the specific use case of *Mural Harbor*, we aim to evaluate the prototype through an in-situ study using semi-structured interviews and questionnaires facilitating the integration of real-world and virtual

elements to create expansive and immersive AR experiences. Future advancements in *Geospatial* tracking will determine its broader applicability and success in AR applications.

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